Medical Terminology

Lectures 1–5 study pack

> **Term Work Exam** · Sunday 25 October 2026 · written · covers Lecture 1 → Lecture 5
> Grading (L1 p8): class activity / periodicals 25 (mid-term / quiz 20 + research 5) · final written 75
> Written exam → spelling, word-structure analysis and meanings matter most.

COLOUR CODE (L4 P2 HIGHLIGHTERS)

prefix root combining vowel suffix

hyper / lip / o / protein / emia = A condition of excessive fat and protein in blood

Amber for the combining vowel is our addition. L2 p7

1. **Foundations** — L1 & L2: definitions, word parts, rules, pronunciation, plurals
2. **Word-part tables** — L3 roots & prefixes · L4 prefixes & suffixes · L5 suffixes
3. **Term analysis** — 73 terms split into parts
4. **Look-alikes** — 25 groups that are easy to mix up
5. **Spelling traps** — 56 words to write correctly
6. **Self-test** with answer key · blank analysis sheet

SOURCE: THE DOCTORS' SLIDES

L1 Introduction: definition, word parts, structural analysis — Dr. Shaimaa Gomaa Ibrahim

L2 Medical term components: combining vowels, rules, pronunciation, plurals — Dr. Shaimaa Gomaa Ibrahim

L3 Roots (physical, time, chemistry, diseases) + prefixes (numbers, size, color, direction, position) — Dr. Shaimaa Gomaa Ibrahim

L4 Prefixes (time/position, disease, degree, negative) + suffixes (specialties, condition, disease) — Made by Assoc. Prof. Amira Ahmed · edited by Dr. Hagar Belal Ahmad

L5 Suffixes (recording/instruments, small, surgery, adjective) — Made by Assoc. Prof. Dr. Amira A. Alqazar · edited by Dr. Hagar Belal Ahmad

Training sheet — Dr. Hagar Belal Ahmad Abo-Zalam

Unofficial student summary — the doctors' slides are the reference.
Prepared by Mohammed Aymen · PharmD Clinical 3 · Fall 2026 · page refs like “L3 p24” point to the PDF page of that lecture · items marked (extra) / (زيادة للفهم) are not on the slides.

1 Foundations L1 · L2

Lecture 1 — Introduction: definition, word parts, structural analysis

Medical terminology — definition L1 p9

- It is the language used to precisely describe the human body including its components, processes, conditions affecting it, and procedures performed upon it.
- Medical terminology is used in the field of medicine and pharmacy.

لغة بتوصف جسم الإنسان بدقة: مكوناته، العمليات اللي بتحصل فيه، الحالات اللي بتأثر عليه، والإجراءات اللي بتتعمل عليه.

Word parts & origin L1 p10

- Most medical terms are a combination of word parts: prefix, root (stem), suffix.
- Most of them come from Latin or Greek words.
- Other languages such as English, German and French.

معظم الكلمات Latin أو Greek.

The word parts L1 p11–12

- Root: the word part that is the core of the word.
- Prefix: a word part attached to the beginning of a word root to modify its meaning.
- Suffix: a word part attached to the end of the word root to modify its meaning.
- Combining vowel: a word part, usually an “o” or “i”.

الـ Root هو قلب الكلمة. الـ prefix في الأول والـ suffix في الآخر، والاتنين بيعدّلوا المعنى.

Defining medical terms through word structural analysis L1 p13, 16–20

- You can define a term according to word structure analysis by: prefix, root, suffix.
- Patterns on the slides: Root + suffix (lipemia) · Prefix + root + suffix (hyperlipemia) · Root + combining vowel + suffix (carcinogen).

عشان تعرّف أي term: قسّمه prefix / root / suffix وهات معنى كل جزء.

Competency (intro slide) L1 p6–7

- It is a set of characteristics including knowledge, skills and attitude. All are integrated within a situation which takes place in a setting.
- Consists of a description of the essential skills, knowledge, attitudes, and behaviors required for effective performance of a real-world task or activity (Mrowicki, 1986).
- الجدارات = Knowledge + Skills + Behaviors + Attitudes · الكفاءات = القدرة على توظيف هذه العناصر في الأداء الفعلي

Lecture 2 — Medical term components: combining vowels, rules, pronunciation, plurals

Combining vowels L2 p4–8

- [o] is the most common combining vowel; [i] is the second most common.
- When a medical term has more than one root, each is joined by a vowel.
- They are used to join: (i) root to root, (ii) root to suffix.
- Why? To ease pronunciation.
- Is their presence mandatory? No — not all medical terms have combining vowels.
- Additional roots: often a medical term is formed of two or more roots (hyper/lip/o/protein/emia).

o الأشهر وبعدها i. بتربط root بـ root أو root بـ suffix، ومش إجبارية.

Formation of medical terms L2 p17

- All medical terms build from the root.
- Prefixes and suffixes are attached to the root to modify its meaning.
- Often two or more roots are linked before being modified.

Rules for building terms L2 p9

Memorizing and spelling basic medical word components are the first steps for learning how to form medical terms. The next step is to construct the words using the following steps (rules).

#	RULE	EXAMPLE
1	A combining vowel is used to join root to root as well as to any suffix beginning with a consonant. o بين root وroot، وقبل أي suffix بيبدأ بـ consonant.	electr/o/cardi/o/gram — electrical record of the heart L2 p10
2	A combining vowel is not used before a suffix that begins with a vowel. الـ suffix بيبدأ بـ vowel؟ يبقى مفيش o.	vas + ectomy → vasectomy — excision of a vessel L2 p12–13
3	If the root ends in a vowel and the suffix begins with the same vowel, drop the final vowel from the root and do not use a combining vowel. نفس الحرف مرتين؟ شيل حرف الـ root.	cardi + itis → carditis — inflammation of heart L2 p14
4	When a prefix ends in a vowel and the root begins with a vowel, the final vowel is dropped from the prefix. هنا اللي بيتشال حرف الـ prefix مش الـ root.	para + enter + al → parenteral — pertaining to alongside of the intestine L2 p15
4b	Most often a combining vowel is inserted between two roots even when the second root begins with a vowel. بين two roots حط o حتى لو التاني بيبدأ بـ vowel.	cardi/o/esophag/eal — pertaining to heart and esophagus L2 p16

Pronunciation at the start of a word

LETTERS	SOUND	EXAMPLES
Ch	k	chemical L2 p18
C (before a, o, u)	k	cavity, colon, cure L2 p18

Plurals (Latin / Greek)

ENDING	SINGULAR	PLURAL
-a → -ae	vertebra bone of the spine	**vertebrae** L2 p20, 22

LETTERS	SOUND	EXAMPLES
C (before e, i)	s	cephalic, cirrhosis L2 p18
Dys	dis	dystrophy L2 p18
Eu	u	euphoria L2 p18
Gn	n	gnathic L2 p18
Ph	f	pharmacy L2 p19
Pn	n	pneumonia L2 p19
Ps	s	pseudo L2 p19
Pt	t	ptosis L2 p19
Rh	r	rheumatic L2 p19
X	z	xenon L2 p19

ENDING	SINGULAR	PLURAL
-en → -ina	lumen central opening	**lumina** L2 p20, 22
-ex, ix → -ices	index indicator	**indices** L2 p20, 22
-is → -es	diagnosis identification of diseases	**diagnoses** L2 p20, 22
-nx (anx, inx, ynx) → -nges	phalanx bone of finger or toe	**phalanges** L2 p23
-on → -a	ganglion mass of nerve tissue	**ganglia** L2 p23
-um → -a	ovum egg	**ova** L2 p23
-us → -i	fungus mold	**fungi** L2 p23

Slide header “Pleural” = Plural.

-graph · -gram · -graphy (picture slide) L2 p11

- Electrocardiograph = the machine (instrument) — -graph
- Electrocardiogram = the record — -gram
- Electrocardiography = the process of recording — -graphy

الماكينة / الورقة / العملية

Lumen (picture slide) L2 p21

- Lumen = central opening (diagrams: a capillary with red blood cells inside the lumen; a cross-section with the lumen in the centre).
- Plural: lumina.

Homework on the slide L2 p24**:** analyse Hypothermia, Oncolysis, Primigravida, Megabladder — answers are in Part 3. No answers on the slide; answers in the analysis list are built from the L3 tables.

2 Word-part tables L3 · L4 · L5

Lecture 3 — Roots (physical, time, chemistry, diseases) + prefixes (numbers, size, color, direction, position)

Diplopia (picture slide) L3 p27
- Healthy: normal vision (one image). Diplopia: double vision (two images).

Bicuspid (picture slide) L3 p28
- Normal tricuspid valve vs bicuspid aortic valve.

Roots pertaining to physical factors L3 p4–9

ROOT	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
aer /o	**air, gas**	*aerobe*	Organism that requires air (oxygen) to live
bar /o	**pressure**	*barotrauma*	Injury (trauma) caused by pressure
chrom /o also chromat /o	**color, stain**	*chromatic*	Pertaining to color
cry /o	**cold**	*cryoprobe*	Instrument used to apply extreme cold to tissues The picture version (p4) adds: apply extreme cold (freeze) to tissues, e.g. to destroy & remove prostate.
electr /o	**electricity**	*electroshock*	Shock produced by electricity
kine also kinesi /o kinet /o	**movement**	*kinesis*	Movement
phot /o	**light**	*photosensitive*	Abnormally sensitive to light
radio	**radiation, X-ray**	*radioactive*	Giving off radiation
son /o	**sound**	*ultrasound*	Pertaining to high frequency sound waves (beyond human hearing)
therm /o	**heat, temperature**	*hypothermia*	Low body temperature

Roots pertaining to time L3 p11

ROOT	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
chron /o	**time**	*synchronous*	Occurring together
noct /i	**night**	*nocturnal*	Occurring at night
nyct /o	**night, darkness**	*nyctophobia*	Fear of night or darkness

Roots pertaining to chemistry L3 p13–16

ROOT	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
adip /o	**fat**	*adiposis*	Accumulation of fat in the body (the process)
amyl /o	**starch**	*amyloid · amylase*	Amyloid: resembling (like) starch · Amylase: enzyme breaks down starch into sugar
azot /o	**nitrogen compound**	*azoturia*	Increased nitrogen compounds in urine
ferr /i also ferr /o	**iron**	*ferric · ferrous*	Ferric: containing iron in oxidized form · Ferrous: containing iron in reduced Slide wording ends at “reduced” (meaning: reduced form).
glyc /o	**sugar, glucose**	*hypoglycemia*	Low blood glucose
kal/	**potassium**	*hyperkalemia*	Excess potassium in the blood
lip /o	**lipid**	*liposuction*	Surgical removal of fat by suction
hydr /o	**water, fluid**	*dehydrate*	To lose water Slide wording: “to loss water”.
natr /i	**sodium**	*hyponatremia*	Decrease of sodium in the blood
prote /o	**protein**	*proteolysis*	Dissolving of protein in blood
steat /o	**fatty**	*steatorrhea*	Fatty diarrhea

Easily recognized roots L3 p17

ROOT	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
oxy	**oxygen**		
calci	**calcium**		

Roots pertaining to diseases L3 p19–20

ROOT	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
carcin /o	**cancer**	*carcinogen*	Substance that produces cancer
lith	**stone**	*lithiasis*	Stone formation
onc /o	**tumor**	*oncolysis*	Destruction (lysis) of a tumor
path /o	**disease**	*pathogenic*	Producing disease
py /o	**pus**	*pyorrhea*	Discharge (rhea) of pus
pyr /o	**fever, fire**	*pyretic*	Pertaining to fever
scler /o	**hard**	*sclerosis*	Hardening of tissue

ROOT	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
tox /o	**poison, toxin**	*toxic*	Poisonous, damaging

Prefixes pertaining to numbers L3 p24–30

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
prim /i	**first**	*primigravida*	Woman pregnant (gravid) for the first time
mon /o	**one**	*monoclonal*	A cell colony (clone) arising from a single cell
uni	**one**	*unicellular*	Organism composed of a single cell
di	**two, twice**	*diplegia*	Double paralysis
diplo	**double**	*diplopia*	Double vision
bi	**two, twice**	*bicuspid*	Having or ending in two points (cuspus) Picture on slide: bicuspid aortic valve vs normal tricuspid valve.
hemi	**half, one side**	*hemithorax*	One side of the chest (thorax) Slide writes “Hemi thorax”; one word: hemithorax.
semi	**half, partial**	*semilunar*	Resembling a half moon
tri	**three**	*triad*	Group of three
quadr /i	**four**	*quadrant*	One fourth of an area
penta	**five**	*pentanoic*	An acid with 5 carbons
multi	**many**	*multiple*	Consisting of many parts
poly	**many, much**	*polymorphic*	Having many shapes (morph/o)

Number prefixes in medical & pharmaceutical terms (chart) L3 p23

Chart slide: it lists examples only, no definitions.

PREFIX	NUMBER	EXAMPLES ON THE CHART
mono-	one	Monotherapy · Monoclonal antibody · Monosaccharide
di-	two	Dicarboxylic acid · Dibasic · Diphenhydramine
tri-	three	Tricyclic antidepressant · Triiodothyronine (T3) · Triglyceride
tetra-	four	Tetracycline · Tetramethyl · Tetrahydrate
penta-	five	Pentamidine · Pentose · Pentapeptide
hexa-	six	Hexachlorophene · Hexose · Hexamethonium
hepta-	seven	Heptavalent · Heptanoic acid · Heptahydrate
octa-	eight	Octanol · Octapeptide · Octahydrate

PREFIX	NUMBER	EXAMPLES ON THE CHART
nona-	nine	Nonavalent · Nonapeptide · Nonanoic acid
deca-	ten	Decamethonium · Decanoic acid · Decapeptide

Prefixes pertaining to size and comparison L3 p32–35

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
eu	**true, good, easy, normal**	*euthanasia*	Easy or painless death
hetero	**other, different, unequal**	*heterosexual*	Pertaining to the opposite sex
homo also homeo	**same, unchanging**	*homothermic*	Maintaining the body temp
iso	**same, equal**	*isograft*	Graft between two genetically identical individuals
macro	**large, abnormally large**	*macroscopic*	Visible with the naked eye (without a microscope)
mega also megalo	**large, abnormally large**	*megabladder*	Permanent enlargement of the bladder
micro	**small**	*microsurgery*	Surgery of extremely small structures under the microscope
neo	**new**	*neonate*	Newborn infant — infant in the first 28 days after birth
normo	**normal**	*normovolemia*	Normal blood volume
ortho	**straight, upright**	*orthosis*	Appliance/device used to correct or prevent deformities
pseudo	**false**	*pseudoparalysis* · *pseudocyesis*	False paralysis · False pregnancy
re	**again, back**	*reflux*	Backward or return flow (reflux esophagitis)

Prefixes pertaining to color L3 p36–37

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
cyan /o	**blue**	*cyanosis*	Bluish discoloration of the skin due to lack of oxygen
erythr /o	**red**	*erythrocyte*	Red blood cell
leuk /o	**white, colorless**	*leukoplakia*	White patches in the mouth
melan /o	**black, dark**	*melanin*	The dark pigment that colors the hair and skin
xanth /o	**yellow**	*xanthoma*	Yellow raised area on the skin

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
icter/	**yellow**	*icterus index*	Index denoting blood concentration of bilirubin
polio also **glauc/o**	**grey**	*polioencephalitis*	Inflammation of the gray matter of the brain Slide writes “Polioencephalities”; correct: Polioencephalitis.
verd/o	**green**	*verdoglobin*	Any of several green compounds derived from hemoglobin or related compounds

Prefixes pertaining to direction L3 p38–39, 41

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
ab	**away from**	*abduct*	To move away from the midline
ad	**toward, near**	*adhesion*	An attachment of parts
dia	**through**	*dialysis*	Separation (-lysis) by passage through a membrane
per	**through**	*percutaneous*	Through the skin
trans	**through**	*transfusion*	Introduction of blood or blood components into blood stream
ex/o	**away from, outside**	*excise*	To cut out
endo	**in, within**	*endoscope*	Device for viewing the inside of a cavity or organ

Prefixes pertaining to position L3 p40

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
dextro	**right**	*dextrocardia*	(Rare condition) Location of the heart in the right side of the chest
levo	**left**	*levorotation*	Rotation towards the left
sinistr/o	**left**	*sinistrad*	Towards the left Slide writes the prefix as “Sinistro/o”.
ecto	**out, outside**	*ectopic*	Outside the normal position (ectopic pregnancy: outside the uterus)

Lecture 4 — Prefixes (time/position, disease, degree, negative) + suffixes (specialties, condition, disease)

Color code for medical terminology L4 p2

- Prefix- (blue): meaning before the root · modifies the meaning · e.g. hyper-, hypo-, anti-, pre-
- -Root- (green): main meaning of the term · usually indicates the organ, system or action · e.g. cardi-, nephro-, hepat-, gastr-
- -Suffix (pink): meaning after the root · indicates condition, procedure, disease, drug class, etc. · e.g. -itis, -emia, -ology, -logy, -ase

Prefix categories (L4) L4 p3

- Time and position
- Disease
- Degree
- Negative prefix

Suffixes L4 p9

- Suffixes are the ending part of medical terms.
- Suffixes tend to bring an action to the medical terms.
- Medical suffixes can be categorized into: specialties, condition of, disease, small, surgery, adjective suffix.

ال suffix آخر الكلمة وبيدّيها action.

Disease — definition L4 p12

- A condition of a person, animal, or plant in which its body or structure is harmed because an organ or part is unable to work as it usually does.

Prefixes pertaining to time and/or position L4 p4

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
ante-	**before**	*antenatal*	Before birth
pre-	**before, in front of**	*predisposing*	Leading to a condition such as a disease (predisposing factors)
pro-	**before, in front of**	*prophylaxis*	Prevention of disease
post-	**after, behind**	*postoperative*	After a surgical operation

Prefixes pertaining to disease L4 p5

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
brady-	**slow**	*bradycardia*	Slow heart rate
tachy-	**rapid**	*tachypnea*	Rapid breathing
dys-	**abnormal, painful, difficult**	*dysfunction*	Abnormal function
mal-	**bad**	*malnutrition*	Poor nutrition
pachy-	**thick**	*pachyderma*	Thickness of the skin
xero-	**dry**	*xerosis*	Dryness of the skin or membranes

Prefixes pertaining to degree L4 p6

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
hyper-	**over, excess, abnormally high, high, increased**	*hypertension*	High blood pressure
hypo-	**under, below, abnormally low, decreased**	*hypoventilation*	Lower amount of air entering lungs
olig /o	**few or deficiency of**	*oligospermia* · *oliguria*	Low number of sperm in the semen · Low urine output Slide spells it “Oligouria”; standard spelling Oliguria (both accepted in quizzes).
pan-	**all**	*panplegia*	Total paralysis
super-	**above or excess**	*superinfection*	Second infection

Negative prefixes L4 p7

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
dis-	**down, removal, separation**	*disinfectant*	To remove infectious organisms
in- also **im-**	**not**	*incontinence* · *imbalance*	Lack of control over excretion (• not moving) · Not being in balance
un-	**not**	*unconscious*	Not responsive or aware or awake
a- also **an-**	**not**	*analgesia*	Pain killer

Anti- L4 p8

Separate picture slide after the negative-prefix table.

PREFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
anti-	**against**	*anti-hypertensive drugs*	Examples on slide: anti-hypertensive drugs (Concor 5 mg) · anti-protozoal drugs (Flagyl 500 mg) · anti-viral drugs (Acyclovir 5%) · antibiotics (Augmentin 1 g) The slide gives no written meaning for anti-; “against” is (extra).

Anti- (picture slide) L4 p8

- Anti-hypertensive drugs (Concor 5 mg)
- Anti-protozoal drugs (Flagyl 500 mg)
- Anti-viral drugs (Acyclovir 5%)
- Antibiotics (Augmentin 1 g)

Suffixes pertaining to specialties L4 p10

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-logy	**study of**	*physiology*	Study of function in a living organism Indicates a FIELD of study
-iatrics	**field of medicine**	*pediatrics*	Study of treatment of children diseases Indicates a FIELD of study
-iatry	**field of medicine**	*psychiatry*	Study of treatment of mental illness Indicates a FIELD of study
-ist	**specialist in a field of study**	*dentist*	Specialist in study and treatment of the teeth and mouth Indicates a PERSON (specialist)
-ian also **-ician**	**specialist in a field of study**	*physician*	Practitioner of medicines Indicates a PERSON (specialist)
-iatrist	**specialist in a field of study**	*podiatrist*	Specialist in study and treatment of the foot Indicates a PERSON (specialist)

Suffixes that mean "condition of" L4 p11

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-ia	**condition of**	*insomnia*	Inability to sleep
-iasis	**condition of**	*amebiasis*	Infection with ameba (parasites)
-ism	**condition of**	*alcoholism*	Chronic dependence or addiction to alcohol
-osis	**condition of**	*neurosis*	Abnormal condition of the neurons
-y	**condition of**	*atony*	Lack of muscle tone

Suffixes pertaining to disease L4 p13–14

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-algia	**pain**	*gastralgia*	Pain in the stomach
-dynia also **-dysnia**	**pain**	*cephalodynia*	Pain in the head (headache) Slide lists "-Dysnia" and "-Dynia"; the example uses -dynia. "-dysnia" is probably a slide slip — learn -dynia.
-cele	**hernia**	*cystocele*	Hernia of the urinary bladder (cyst/o)
-clasis also **-clasia**	**breaking**	*karyoclasis (karyolysis)* · *cytoclasis*	Breaking of a cell nucleus · Breaking or destruction of a cell
-itis	**inflammation**	*colitis*	Inflammation of the colon
-megaly	**enlargement**	*hepatomegaly*	Enlargement of the liver (hepat/o)

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-oma	**tumor**	*lipoma*	Fatty tumor (lip = fat, oma = tumor)
-pathy	**any disease of**	*neuropathy*	Any disease of the nervous system (neuro = nerve, pathy = disease)
-rrhage also **-rrhagia**	**bursting, profuse flow**	*hemorrhage*	Profuse flow of blood Slide header writes "-Rhage, -Rhagia"; in the word it is always double r: hemorrhage.
-rrhea	**flow, discharge**	*diarrhea*	Frequent passage of watery stool Slide header writes "-Rhea". Its word split "diarr = through, hea = flow" is a slide slip: dia = through, -rrhea = flow.
-rrhexis	**rupture**	*amniorrhexis*	Rupture of the amniotic sac (bag of waters) Slide writes "Reputure" / "Rapture" → Rupture; header "-Rhexis" → -rrhexis.
-schisis	**fissure, splitting**	*retinoschisis*	Splitting of the retina of the eye

Lecture 5 — Suffixes (recording/instruments, small, surgery, adjective)

Adjective suffix — definition L5 p7

- A suffix added to a root word to form an adjective.

Suffixes pertaining to disease — recording, measuring & viewing (-graph, -gram, -meter, -scope ...) L5 p3

Slide title is "Suffix pertaining to disease" (L5 p3); the content is instruments/recording.

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-graph	**instrument for recording data**	*cardiograph*	Instrument used to record the electrical signal from the heart to check for different heart conditions
-graphy	**recording data**	*roentgenography, radiography*	Obtaining pictures with roentgen rays (x-rays)
-gram	**a record of data**	*sonogram*	Record obtained by ultrasonography
-meter	**instrument for measuring**	*spirometer*	Instrument for measuring ventilation and breathing
-metry	**measurement**	*optometry*	Measurement of vision (opt/o) for glasses
-scope	**instrument for viewing or for examining**	*cystoscope*	Endoscope used to examine the urinary bladder
-scopy	**examination**	*laparoscopy*	Examination of the abdomen through the abdominal wall (lapar/o)

Suffixes that mean “small” L5 p4

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-cle	**small**	*ventricle*	Small belly or pouch — a cavity
-ole	**small**	*bronchiole*	Small airway
-ula	**small**	*macula*	Small spot
-ule	**small**	*pustule*	Small pimple

Suffixes pertaining to surgery L5 p5–6

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-centesis	**puncture or tap**	*celiocentesis*	Puncture of the abdomen
-ectomy	**excision, surgical removal**	*tonsillectomy*	Excision of the tonsils (tonsill/o)
-desis	**binding, fusion**	*pleurodesis*	Binding the pleural membranes (around the lungs)
-pexy	**surgical fixation**	*uteropexy*	Surgical fixation of the uterus (uter/o)
-plasty	**plastic repair, reconstruction**	*septoplasty*	Plastic repair of the nasal septum (partition)
-rrhaphy	**surgical repair, suture**	*herniorrhaphy*	Surgical repair of the hernia (herni/o)
-stomy	**surgical creation of an opening**	*colostomy*	Creation of an opening into the colon
-tome	**instrument for incision (cutting)**	*dermatome*	Instrument for cutting the skin
-tomy	**incision, cutting**	*laparotomy*	Surgical incision of the abdominal wall
-tripsy	**crushing**	*lithotripsy*	Crushing of a stone

Adjective suffixes L5 p8

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-ac	**pertaining to**	*cardiac*	Pertaining to the heart
-al	**pertaining to**	*skeletal*	Pertaining to the skeleton
-ar	**pertaining to**	*muscular*	Pertaining to the muscles
-ary	**pertaining to**	*urinary*	Pertaining to the urine
-ic	**pertaining to**	*pelvic*	Pertaining to the pelvis
-ical	**pertaining to**	*surgical*	Pertaining to surgery
-ile	**pertaining to**	*febrile*	Pertaining to fever

SUFFIX	MEANING	EXAMPLE	DEFINITION OF THE EXAMPLE
-ous	**pertaining to**	*venous*	Pertaining to a vein
-ive	**pertaining to**	*effective*	Pertaining to having an intended effect Slide wording: “having and intended effect”.
-form	**similar to**	*epileptiform*	Like or resembling epilepsy
-oid	**similar to**	*mucoid*	Like or resembling mucus

3 Term analysis prefix / root / vowel / suffix

Write it like the training sheet: term → structure → meaning of each part → meaning of the term. ★ = worked on the slide. * = part meaning is extra (not written on the slide).

TERM & STRUCTURE	PARTS	MEANING OF THE TERM
Carcinogen ★ carcin / o / gen L1 p13	carcin cancer gen producing	Substance that promotes the formation of cancer
Pericarditis ★ peri / card / itis L1 p14	peri around card heart itis inflammation	Inflammation around the heart
Lipemia ★ lip / emia L1 p15	lip fat emia blood condition	A condition of fat in blood
Hyperlipemia ★ hyper / lip / emia L1 p19	hyper excessive lip fat emia blood condition	A condition of excessive fat in blood
Cardiac ★ cardi / ac L1 p20	cardi heart ac pertaining to / related to	Pertaining to heart
Hyperlipoproteinemia ★ hyper / lip / o / protein / emia L2 p7	hyper excessive lip fat protein protein emia blood condition	A condition of excessive fat and protein in blood
Electrocardiogram ★ electr / o / cardi / o / gram L2 p10	electr electric cardi heart gram record	Electrical record of the heart
Vasectomy ★ vas / ectomy L2 p13	vas vessel ectomy excision (استئصال)	Excision of a vessel Rule 2: no combining vowel before a suffix that begins with a vowel.
Carditis ★ card(i) / itis L2 p14	card(i) heart itis inflammation	Inflammation of heart Rule 3: root ends in i, suffix begins with i → drop the root's i.
Parenteral ★ par(a) / enter / al L2 p15	par(a) alongside of enter intestine al pertaining to	Pertaining to alongside of the intestine Rule 4: prefix ends in a vowel, root begins with a vowel → drop the prefix's a.
Cardioesophageal ★ cardi / o / esophag / eal L2 p16	cardi heart esophag esophagus eal pertaining to	Pertaining to heart and esophagus Rule 4b: combining vowel between two roots even when the second root begins with a vowel.
Hypothermia hypo / therm / ia L2 p24	hypo under, below, abnormally low therm heat, temperature ia condition of	Low body temperature

TERM & STRUCTURE	PARTS	MEANING OF THE TERM
Oncolysis onc / o / lysis L2 p24	onc tumor lysis destruction	Destruction (lysis) of a tumor
Primigravida primi / gravid / a L2 p24	primi first gravid pregnant a —	Woman pregnant (gravid) for the first time The final -a is a noun ending; the slides don't give it a meaning.
Megabladder mega / bladder L2 p24	mega large, abnormally large bladder bladder	Permanent enlargement of the bladder No suffix — prefix + root only.
Hypoglycemia hypo / glyc / emia L3 p14	hypo under, below, decreased glyc sugar, glucose emia blood condition	Low blood glucose
Hyperkalemia hyper / kal / emia L3 p15	hyper over, excess, increased kal potassium emia blood condition	Excess potassium in the blood
Hyponatremia hypo / natr / emia L3 p16	hypo under, below, decreased natr sodium emia blood condition	Decrease of sodium in the blood
Steatorrhea steat / o / rrhea L3 p16	steat fatty rrhea flow, discharge	Fatty diarrhea
Azoturia azot / uria L3 p13	azot nitrogen compound uria urine (condition)	Increased nitrogen compounds in urine On the slide “uria” and “urine” are highlighted in red.
Pyorrhea py / o / rrhea L3 p20	py pus rrhea discharge	Discharge (rhea) of pus
Cyanosis cyan / osis L3 p36	cyan blue osis condition of	Bluish discoloration of the skin due to lack of oxygen
Erythrocyte erythr / o / cyte L3 p36	erythr red cyte cell	Red blood cell
Polioencephalitis polio / encephal / itis L3 p37	polio grey encephal brain itis inflammation	Inflammation of the gray matter of the brain
Diplopia dipl(o) / opia L3 p26	dipl(o) double opia vision	Double vision The o of diplo is dropped before opia.
Hemithorax hemi / thorax L3 p29	hemi half, one side thorax chest	One side of the chest (thorax)
Normovolemia normo / vol / emia L3 p34	normo normal vol volume emia blood condition	Normal blood volume

TERM & STRUCTURE	PARTS	MEANING OF THE TERM
Percutaneous per / cutane / ous L3 p39	per through cutane skin ous pertaining to	Through the skin
Endoscope endo / scope L3 p41	endo in, within scope instrument for viewing or examining	Device for viewing the inside of a cavity or organ
Dextrocardia dextro / card / ia L3 p40	dextro right card heart ia condition of	Location of the heart in the right side of the chest
Bradycardia brady / card / ia L4 p5	brady slow card heart ia condition of	Slow heart rate
Tachypnea tachy / pnea L4 p5	tachy rapid pnea breathing	Rapid breathing
Xerosis xer(o) / osis L4 p5	xer(o) dry osis condition of	Dryness of the skin or membranes
Pachyderma pachy / derma L4 p5	pachy thick derma skin	Thickness of the skin
Oliguria olig / uria L4 p6	olig few or deficiency of uria urine (condition)	Low urine output Slide spells it “Oligouria”.
Panplegia pan / pleg / ia L4 p6	pan all pleg paralysis ia condition of	Total paralysis
Gastralgia gastr / algia L4 p13	gastr stomach algia pain	Pain in the stomach
Cephalodynia cephal / o / dynia L4 p13	cephal head dynia pain	Pain in the head (headache)
Cystocele cyst / o / cele L4 p13	cyst urinary bladder cele hernia	Hernia of the urinary bladder
Colitis col / itis L4 p13	col colon itis inflammation	Inflammation of the colon
Hepatomegaly hepat / o / megaly L4 p13	hepat liver megaly enlargement	Enlargement of the liver
Lipoma ★ lip / oma L4 p14	lip fat oma tumor	Fatty tumor
Neuropathy ★ neur / o / pathy L4 p14	neur nerve pathy any disease of	Any disease of the nervous system
Hemorrhage ★ hem / o / rrhage L4 p14	hem blood rrhage bursting, profuse flow	Profuse flow of blood Slide split: hemor = blood, rhage = flow.

TERM & STRUCTURE	PARTS	MEANING OF THE TERM
Amniorrhexis ★ amni / o / rrhexis L4 p14	amni amniotic sac rrhexis rupture	Rupture of the amniotic sac (bag of waters)
Retinoschisis ★ retin / o / schisis L4 p14	retin retina schisis splitting	Splitting of the retina of the eye
Cardiograph cardi / o / graph L5 p3	cardi heart graph instrument for recording data	Instrument used to record the electrical signal from the heart to check for different heart conditions
Sonogram son / o / gram L5 p3	son sound gram a record of data	Record obtained by ultrasonography
Optometry opt / o / metry L5 p3	opt vision metry measurement	Measurement of vision for glasses
Cystoscope cyst / o / scope L5 p3	cyst urinary bladder scope instrument for viewing or examining	Endoscope used to examine the urinary bladder
Laparoscopy lapar / o / scopy L5 p3	lapar abdomen scopy examination	Examination of the abdomen through the abdominal wall
Bronchiole bronchi / ole L5 p4	bronchi airway (bronchus) ole small	Small airway
Celiocentesis celi / o / centesis L5 p5	celi abdomen centesis puncture or tap	Puncture of the abdomen
Tonsillectomy tonsill / ectomy L5 p5	tonsill tonsils ectomy excision, surgical removal	Excision of the tonsils
Pleurodesis pleur / o / desis L5 p5	pleur pleura (membranes around the lungs) desis binding, fusion	Binding the pleural membranes (around the lungs)
Uteropexy uter / o / pexy L5 p5	uter uterus pexy surgical fixation	Surgical fixation of the uterus
Septoplasty sept / o / plasty L5 p5	sept septum (partition) plasty plastic repair, reconstruction	Plastic repair of the nasal septum (partition)
Herniorrhaphy herni / o / rrhaphy L5 p5	herni hernia rrhaphy surgical repair, suture	Surgical repair of the hernia
Colostomy col / o / stomy L5 p6	col colon stomy surgical creation of an opening	Creation of an opening into the colon
Dermatome derma / tome L5 p6	derma skin tome instrument for incision (cutting)	Instrument for cutting the skin

TERM & STRUCTURE	PARTS	MEANING OF THE TERM
Laparotomy lapar / o / tomy L5 p6	lapar abdomen tomy incision, cutting	Surgical incision of the abdominal wall
Lithotripsy lith / o / tripsy L5 p6	lith stone tripsy crushing	Crushing of a stone
Diarrhea ★ dia / rrhea L4 p14	dia through rrhea flow, discharge	Frequent passage of watery stool Slide split “diarr = through, hea = flow” (word printed “Diarrrhea”) is a slip: the same slide table gives -rrhea = flow, so dia = through + -rrhea = flow.
Antenatal ante / nat / al L4 p4	ante before nat birth al pertaining to	Before birth
Postoperative post / operat / ive L4 p4	post after, behind operat (surgical) operation ive pertaining to	After a surgical operation
Hypertension hyper / tens / ion L4 p6	hyper over, excess, abnormally high, high, increased tens (blood) pressure ion —	High blood pressure
Superinfection super / infect / ion L4 p6	super above or excess infect infection ion —	Second infection
Imbalance im / balance L4 p7	im not balance balance	Not being in balance
Analgesia an / alges / ia L4 p7	an not alges pain ia —	Pain killer
Ventricle ventri / cle L5 p4	ventri belly, pouch cle small	Small belly or pouch — a cavity
Pustule pust / ule L5 p4	pust pimple ule small	Small pimple
Mucoid muc / oid L5 p8	muc mucus oid similar to (like, resembling)	Like or resembling mucus
Febrile febr / ile L5 p8	febr fever ile pertaining to	Pertaining to fever

4 Look-alikes don't mix them up

-graph · -graphy · -gram L5 p3 · L2 p11

-graph	instrument for recording data	*cardiograph*
-graphy	recording data	*radiography*
-gram	a record of data	*sonogram*

الماكينة ← العملية ← الورقة اللي طالعة: gram ← graphy ← graph (زيادة للفهم)

-meter · -metry L5 p3

-meter	instrument for measuring	*spirometer*
-metry	measurement	*optometry*

اللي بيخلص بـ y هو العملية: metry, scopy, graphy, tomy (زيادة للفهم)

-scope · -scopy L5 p3

-scope	instrument for viewing or for examining	*cystoscope, endoscope*
-scopy	examination	*laparoscopy*

scope = الأداة، scopy = الفحص نفسه

-tome · -tomy · -ectomy · -stomy L5 p5–6

-tome	instrument for incision (cutting)	*dermatome*
-tomy	incision, cutting	*laparotomy*
-ectomy	excision, surgical removal	*tonsillectomy*
-stomy	surgical creation of an opening	*colostomy*

ec = out: ectomy بتشيل العضو. stomy = فتحة (زيادة للفهم)

-rrhage · -rrhea · -rrhexis · -rrhaphy L4 p14 · L5 p5

-rrhage, -rrhagia	bursting, profuse flow	*hemorrhage*
-rrhea	flow, discharge	*diarrhea*
-rrhexis	rupture	*amniorrhexis*
-rrhaphy	surgical repair, suture	*herniorrhaphy*

rrhaphy بس هي اللي surgery (L5)، الباقي disease (L4)

-pexy · -desis · -plasty L5 p5

-pexy	surgical fixation	*uteropexy*
-desis	binding, fusion	*pleurodesis*
-plasty	plastic repair, reconstruction	*septoplasty*

-algia · -dynia L4 p13

-algia	pain	*gastralgia*
-dynia	pain	*cephalodynia*

الاتنين = pain

-iatry · -iatrics · -iatrist L4 p10

-iatry	field of medicine	*psychiatry*
-iatrics	field of medicine	*pediatrics*
-iatrist	specialist (person)	*podiatrist*

ist- في الآخر = شخص

Hyper · Hypo · Super · Olig/o · Pan L4 p6

hyper-	over, excess, abnormally high	*hypertension*
hypo-	under, below, abnormally low	*hypoventilation*
super-	above or excess	*superinfection*
olig/o	few or deficiency of	*oliguria*
pan-	all	*panplegia*

Ante · Pre · Pro · Post L4 p4

ante-	before	*antenatal*
pre-	before, in front of	*predisposing*
pro-	before, in front of	*prophylaxis*
post-	after, behind	*postoperative*

Dia · Per · Trans L3 p38–39

dia	through	*dialysis*
per	through	*percutaneous*
trans	through	*transfusion*

through = التلاتة

Ab · Ad · Ex/o · Endo · Ecto L3 p38–41

ab	away from	*abduct*
ad	toward, near	*adhesion*
ex /o	away from, outside	*excise*
endo	in, within	*endoscope*
ecto	out, outside	*ectopic*

Ab = Away، Ad = aDd (يقرّب) (زيادة للفهم)

Dextro · Levo · Sinistr/o L3 p40

dextro	right	*dextrocardia*
levo	left	*levorotation*
sinistr /o	left	*sinistrad*

Mon/o · Uni · Prim/i L3 p24–25

mon /o	one	*monoclonal*
uni	one	*unicellular*
prim /i	first	*primigravida*

prim = first مش one

Di · Diplo · Bi L3 p26

di	two, twice	*diplegia*
diplo	double	*diplopia*
bi	two, twice	*bicuspid*

Hemi · Semi L3 p29

hemi	half, one side	*hemithorax*
semi	half, partial	*semilunar*

Macro · Mega · Micro L3 p33

macro	large, abnormally large	*macroscopic*
mega, megalo	large, abnormally large	*megabladder*
micro	small	*microsurgery*

Homo · Iso · Hetero L3 p32

homo, homeo	same, unchanging	*homothermic*
iso	same, equal	*isograft*
hetero	other, different, unequal	*heterosexual*

Noct/i · Nyct/o · Chron/o L3 p11

noct /i	night	*nocturnal*
nyct /o	night, darkness	*nyctophobia*
chron /o	time	*synchronous*

Py/o · Pyr/o L3 p20

py /o	pus	*pyorrhea*
pyr /o	fever, fire	*pyretic*

pyR = fiRe (زيادة للفهم)

Adip/o · Lip/o · Steat/o L3 p13–16

adip /o	fat	*adiposis*
lip /o	lipid	*liposuction*
steat /o	fatty	*steatorrhea*

Kal · Natr/i · Calci · Ferr/o L3 p14–17

kal/	potassium	*hyperkalemia*
natr /i	sodium	*hyponatremia*
calci	calcium	—
ferr/i, ferr /o	iron	*ferric, ferrous*

K = Kalium, Na = Natrium (زيادة للفهم)

Yellow · Grey · Green L3 p37

xanth/o, icter/	yellow	*xanthoma, icterus index*
polio, glauc **/o**	grey	*polioencephalitis*
verd **/o**	green	*verdoglobin*

Small: -cle · -ole · -ula · -ule L5 p4

-cle	small	*ventricle*
-ole	small	*bronchiole*
-ula	small	*macula*
-ule	small	*pustule*

small = الأربعة

Pertaining to vs similar to L5 p8

-ac, -al, -ar, -ary, -ic, -ical, -ile, -ous, -ive	pertaining to	*cardiac … effective*
-form, -oid	similar to (like, resembling)	*epileptiform, mucoid*

5 Spelling traps written exam

Write the left column exactly (American spelling, as on the slides). “Say” is a pronunciation guide (extra).

CORRECT	DON'T WRITE	SAY (EXTRA)
diarrhea L4 p14	~~diarhea · diarrhoea~~ (British spelling — use the slide spelling)	dye-uh-REE-uh
hemorrhage L4 p14	~~hemorage · hemmorhage~~	HEM-or-ij
amniorrhexis L4 p14	~~amniorexis · amniorhexis~~	am-nee-or-REK-sis
herniorrhaphy L5 p5	~~herniorraphy · herniorhaphy~~	her-nee-OR-uh-fee
steatorrhea L3 p16	~~steatorhea · steatorrea~~	stee-at-oh-REE-uh
pyorrhea L3 p20	~~pyorhea · pyrorrhea~~	pye-oh-REE-uh
tonsillectomy L5 p5	~~tonsilectomy~~	ton-sil-LEK-tuh-mee
oliguria L4 p6	oligouria (slide spelling — accepted)	ol-ig-YOO-ree-uh
parenteral L2 p15	~~paraenteral · parentral~~	par-EN-ter-al
carditis L2 p14	~~cardiitis · cardioitis~~	kar-DYE-tis
cardioesophageal L2 p16	~~cardiesophageal · cardesophageal~~	KAR-dee-oh-ee-sof-uh-JEE-al
electrocardiogram L2 p10	~~electrcardiogram · electrocardiagram~~	ee-LEK-troh-KAR-dee-oh-gram
vasectomy L2 p13	~~vasoectomy~~	vas-EK-tuh-mee
hyperlipoproteinemia L2 p7	~~hyperlipproteinemia~~	HY-per-LIP-oh-PRO-teen-EE-mee-uh
pediatrics L4 p10	~~pedeatrics · paediatrics~~ (British spelling — use the slide spelling)	pee-dee-AT-riks
psychiatry L4 p10	~~sychiatry · psychiatery~~	sy-KY-uh-tree
podiatrist L4 p10	~~podiatrest~~	poh-DYE-uh-trist
polioencephalitis L3 p37	~~polioencephalities~~ (slide typo)	POH-lee-oh-en-SEF-uh-LYE-tis
hemithorax L3 p29	~~hemi thorax~~ (slide writes two words)	hem-ee-THOR-aks
nyctophobia L3 p11	~~nictophobia · nyctofobia~~	nik-toh-FOH-bee-uh
synchronous L3 p11	~~sinchronous · synchronus~~	SIN-kruh-nus
euthanasia L3 p32	~~euthenasia · uthanasia~~	yoo-thuh-NAY-zhuh
pseudocyesis L3 p35	~~psudocyesis · pseudocysis~~	SOO-doh-sye-EE-sis
xanthoma L3 p37	~~zanthoma · xantoma~~	zan-THOH-muh

CORRECT	DON'T WRITE	SAY (EXTRA)
leukoplakia L3 p36	~~leukoplasia · leucoplakia~~ (British spelling — use the slide spelling)	loo-koh-PLAY-kee-uh
erythrocyte L3 p36	~~erythrosyte · erithrocyte~~	eh-RITH-roh-site
cyanosis L3 p36	~~cyanosys · cianosis~~	sye-uh-NOH-sis
icterus L3 p37	~~ictirus · ecterus~~	IK-ter-us
vertebrae L2 p20	~~vertebras · vertebrea~~	VER-tuh-bray
phalanges L2 p23	~~phalanxes · phalanxs~~	fuh-LAN-jeez
ganglia L2 p23	~~ganglii · ganglions~~ (English plural — the slide wants Latin ganglia)	GANG-lee-uh
lumina L2 p20	~~lumena · lumens~~ (English plural — the slide wants Latin lumina)	LOO-min-uh
indices L2 p20	~~indicies · indexes~~ (English plural — the slide wants Latin indices)	IN-dih-seez
diagnoses L2 p20	~~diagnosises~~	dye-ag-NOH-seez
primigravida L3 p25	~~primagravida · primigravidia~~	pry-mih-GRAV-ih-duh
tachypnea L4 p5	~~tachypnia · tachypnoea~~ (British spelling — use the slide spelling)	tak-ip-NEE-uh
pachyderma L4 p5	~~pachiderma · pakyderma~~	pak-ee-DER-muh
hyponatremia L3 p16	~~hyponatrimia · hyponatraemia~~ (British spelling — use the slide spelling)	HY-poh-nuh-TREE-mee-uh
hyperkalemia L3 p15	~~hypercalemia · hyperkalaemia~~ (British spelling — use the slide spelling)	HY-per-kuh-LEE-mee-uh
prophylaxis L4 p4	~~profylaxis · prophylaxes~~	proh-fuh-LAK-sis
xerosis L4 p5	~~zerosis · xerosys~~	zih-ROH-sis
cephalodynia L4 p13	~~cephalodinia · kephalodynia~~	SEF-uh-loh-DIN-ee-uh
retinoschisis L4 p14	~~retinoshisis · retinoschysis~~	RET-in-OS-kih-sis
karyoclasis L4 p13	~~kariyoclasis · karioclasis~~	kar-ee-OK-luh-sis
amebiasis L4 p11	~~ameobiasis · amoebiasis~~ (British spelling — use the slide spelling)	am-ee-BYE-uh-sis
celiocentesis L5 p5	~~celiocentisis · coeliocentesis~~ (British spelling — use the slide spelling)	SEE-lee-oh-sen-TEE-sis
pleurodesis L5 p5	~~pleurodisis · pleuradesis~~	ploo-ROD-uh-sis
lithotripsy L5 p6	~~lithotripsey · lithotrypsy~~	LITH-oh-trip-see
laparoscopy L5 p3	~~laproscopy · laparascopy~~	lap-uh-ROS-kuh-pee
roentgenography L5 p3	~~rontgenography · roentgenograpy~~	RENT-gen-OG-ruh-fee

CORRECT	DON'T WRITE	SAY (EXTRA)
dextrocardia L3 p40	~~dextracardia · dextrocardya~~	dek-stroh-KAR-dee-uh
sinistrad L3 p40	~~sinestrad · sinistrid~~	SIN-is-trad
ptosis L2 p19	~~tosis · ptoses~~	TOH-sis
pneumonia L2 p19	~~neumonia · pnuemonia~~	noo-MOH-nee-uh
gnathic L2 p18	~~nathic · gnatic~~	NATH-ik
rheumatic L2 p19	~~reumatic · rhuematic~~	roo-MAT-ik

6 Self-test 70 questions · answers at the end

More (484 questions, typed answers, flashcards) on the website. The more interactive way to drill.

A. Choose the correct answer

1. The root “aer/o” means: L3 p4–9
   a) cold
   b) air, gas
   c) movement
   d) radiation, X-ray
2. The root “kine” means: L3 p4–9
   a) light
   b) pressure
   c) movement
   d) cold
3. The root “chron/o” means: L3 p11
   a) sugar, glucose
   b) time
   c) cancer
   d) night, darkness
4. The root “azot/o” means: L3 p13–16
   a) starch
   b) nitrogen compound
   c) potassium
   d) fat
5. The root “hydr/o” means: L3 p13–16
   a) potassium
   b) water, fluid
   c) nitrogen compound
   d) sodium
6. The root “lith” means: L3 p19–20
   a) stone
   b) disease
   c) tumor
   d) poison, toxin
7. Which root means “poison, toxin”? L3 p19–20
   a) py/o
   b) tox/o
   c) lith
   d) onc/o
8. “Having or ending in two points (cuspus)” is the definition of: L3 p24–30
   a) bicuspid
   b) hemithorax
   c) quadrant
   d) multiple
9. “An acid with 5 carbons” is the definition of: L3 p24–30
   a) multiple
   b) pentanoic
   c) unicellular
   d) primigravida
10. “Maintaining the body temp” is the definition of: L3 p32–35
    a) homothermic
    b) megabladder
    c) normovolemia
    d) reflux
11. The prefix “normo” means: L3 p32–35
    a) other, different, unequal
    b) normal
    c) large, abnormally large
    d) false
12. “White patches in the mouth” is the definition of: L3 p36–37
    a) leukoplakia
    b) verdoglobin
    c) cyanosis
    d) icterus index

13. Which prefix means “green”? L3 p36–37
    a) leuk/o
    b) melan/o
    c) icter/
    d) verd/o
14. The prefix “ex/o” means: L3 p38–39, 41
    a) away from, outside
    b) toward, near
    c) through
    d) in, within
15. Which prefix means “before”? L4 p4
    a) tachy-
    b) uni
    c) ante-
    d) post-
16. Which prefix means “rapid”? L4 p5
    a) mal-
    b) dys-
    c) pachy-
    d) tachy-
17. “Lower amount of air entering lungs” is the definition of: L4 p6
    a) oligospermia
    b) hypertension
    c) panplegia
    d) hypoventilation
18. “Lack of control over excretion (• not moving)” is the definition of: L4 p7
    a) polymorphic
    b) percutaneous
    c) diplegia
    d) incontinence
19. “Practitioner of medicines” is the definition of: L4 p10
    a) septoplasty
    b) physician
    c) effective
    d) colitis
20. “Pain in the head (headache)” is the definition of: L4 p13–14
    a) retinoschisis
    b) karyoclasis
    c) cephalodynia
    d) lipoma
21. “Fatty tumor (lip = fat, oma = tumor)” is the definition of: L4 p13–14
    a) gastralgia
    b) lipoma
    c) retinoschisis
    d) neuropathy
22. The suffix “-graph” means: L5 p3
    a) examination
    b) measurement
    c) incision, cutting
    d) instrument for recording data
23. The suffix “-scope” means: L5 p3
    a) measurement
    b) a record of data
    c) pertaining to
    d) instrument for viewing or for examining
24. “Puncture of the abdomen” is the definition of: L5 p5–6
    a) celiocentesis
    b) septoplasty
    c) lithotripsy
    d) dermatome
25. “Surgical repair of the hernia (herni/o)” is the definition of: L5 p5–6
    a) lithotripsy
    b) pleurodesis
    c) herniorrhaphy
    d) laparotomy

26. “Pertaining to the heart” is the definition of: L5 p8
    a) cardiac
    b) neurosis
    c) mucoid
    d) hemorrhage
27. The suffix “-form” means: L5 p8
    a) small
    b) similar to
    c) pertaining to
    d) pain
28. In the number-prefix chart, “penta-” means: L3 p23
    a) ten
    b) eight
    c) one
    d) five
29. According to the slide (Latin plural), the plural of “vertebra” is: L2 p20, 22
    a) vertebrae
    b) vertebri
    c) vertebras
    d) vertebrina
30. According to the slide (Latin plural), the plural of “ovum” is: L2 p23
    a) ovae
    b) ovums
    c) ova
    d) ovi
31. At the start of a word, “Eu” is pronounced like: L2 p18
    a) u
    b) t
    c) n
    d) r
32. At the start of a word, “X” is pronounced like: L2 p19
    a) k
    b) s
    c) z
    d) r
33. The final written exam is worth: L1 p8
    a) 75 marks
    b) 50 marks
    c) 25 marks
    d) 20 marks
34. cardi + itis = carditis. Which rule explains it? L2 p14
    a) Root ends in a vowel and suffix begins with the same vowel → drop the root's final vowel
    b) A combining vowel joins root to root
    c) Drop the final vowel of the prefix
    d) Combining vowel is inserted between two roots
35. Which is NOT one of the suffix categories in the slides? L4 p9
    a) Degree
    b) Specialties
    c) Surgery
    d) Adjective suffix
36. An adjective suffix is: L5 p7
    a) A suffix added to a root word to form an adjective
    b) A suffix that means small
    c) A suffix used in surgery
    d) A prefix that means not

B. True or false — correct the false ones

37. “bar/o” means potassium. L3 p4–9
38. “adip/o” means fat. L3 p13–16
39. “py/o” means pus. L3 p19–20
40. “multi” means same, equal. L3 p24–30
41. “erythr/o” means red. L3 p36–37

42. “levo” means left. L3 p40
43. “olig/o” means through. L4 p6
44. “-ism” means any disease of. L4 p11
45. “-graph” means instrument for recording data. L5 p3
46. “-plasty” means surgical fixation. L5 p5–6
47. “-form” means specialist in a field of study. L5 p8
48. The suffix is the core of the medical word. L1 p11
49. Levo and sinistro both mean left. L3 p40
50. Isograft is a graft between two genetically identical individuals. L3 p32

C. Write the term / plural (spelling counts)

51. Write the medical term (aer/o = air, gas): “Organism that requires air (oxygen) to live” L3 p4–9

52. Write the medical term (chron/o = time): “Occurring together” L3 p11

53. Write the medical term (hydr/o = water, fluid): “To lose water” L3 p13–16

54. Write the medical term (scler/o = hard): “Hardening of tissue” L3 p19–20

55. Write the medical term (bi = two, twice): “Having or ending in two points (cuspus)” L3 p24–30

56. Write the medical term (macro = large, abnormally large): “Visible with the naked eye (without a microscope)” L3 p32–35

57. Write the medical term (cyan/o = blue): “Bluish discoloration of the skin due to lack of oxygen” L3 p36–37

58. Write the medical term (dia = through): “Separation (-lysis) by passage through a membrane” L3 p38–39, 41

59. Write the medical term (sinistr/o = left): “Towards the left” L3 p40

60. Write the medical term (dys- = abnormal, painful, difficult): “Abnormal function” L4 p5

61. Write the medical term (a-, an- = not): “Pain killer” L4 p7

62. Write the medical term (-ian, -ician = specialist in a field of study): “Practitioner of medicines” L4 p10

63. Write the medical term (-osis = condition of): “Abnormal condition of the neurons” L4 p11

64. Write the medical term (-itis = inflammation): “Inflammation of the colon” L4 p13–14

65. Write the medical term (-gram = a record of data): “Record obtained by ultrasonography” L5 p3

66. Write the medical term (-ectomy = excision, surgical removal): “Excision of the tonsils (tonsill/o)” L5 p5–6

67. Write the medical term (-ac = pertaining to): “Pertaining to the heart” L5 p8

68. Write the medical term (-ical = pertaining to): “Pertaining to surgery” L5 p8

69. Write the medical term (-oid = similar to): “Like or resembling mucus” L5 p8

70. According to the slide (Latin plural), write the plural of “diagnosis” (identification of diseases). L2 p20, 22

Answer key

1. **b** air, gas
2. **c** movement
3. **b** time
4. **b** nitrogen compound
5. **b** water, fluid
6. **a** stone
7. **b** tox/o
8. **a** bicuspid
9. **b** pentanoic
10. **a** homothermic
11. **b** normal
12. **a** leukoplakia
13. **d** verd/o
14. **a** away from, outside
15. **c** ante-
16. **d** tachy-
17. **d** hypoventilation
18. **d** incontinence
19. **b** physician
20. **c** cephalodynia
21. **b** lipoma
22. **d** instrument for recording data
23. **d** instrument for viewing or for examining
24. **a** celiocentesis
25. **c** herniorrhaphy
26. **a** cardiac
27. **b** similar to
28. **d** five
29. **a** vertebrae
30. **c** ova
31. **a** u
32. **c** z
33. **a** 75 marks
34. **a** Root ends in a vowel and suffix begins with the same vowel → drop the root's final vowel
35. **a** Degree
36. **a** A suffix added to a root word to form an adjective
37. **False** — “bar/o” means pressure; “kal/” means potassium.
38. **True**
39. **True**
40. **False** — “multi” means many; “iso” means same, equal.
41. **True**
42. **True**
43. **False** — “olig/o” means few or deficiency of; “trans” means through.
44. **False** — “-ism” means condition of; “-pathy” means any disease of.
45. **True**
46. **False** — “-plasty” means plastic repair, reconstruction; “-pexy” means surgical fixation.
47. **False** — “-form” means similar to; “-iatrist” means specialist in a field of study.
48. **False** — The ROOT is the core of the word.
49. **True**
50. **True**
51. **aerobe**
52. **synchronous**
53. **dehydrate**
54. **sclerosis**
55. **bicuspid**
56. **macroscopic**
57. **cyanosis**
58. **dialysis**
59. **sinistrad**
60. **dysfunction**
61. **analgesia**
62. **physician**
63. **neurosis**
64. **colitis**
65. **sonogram**
66. **tonsillectomy**
67. **cardiac**
68. **surgical**
69. **mucoid**
70. **diagnoses**

Analysis practice sheet

Same layout as the training sheet by Dr. Hagar Belal Ahmad Abo-Zalam. Try the homework terms and any term from Part 3, then check.

MEDICAL TERM	TERM STRUCTURE			MEANING OF THE MEDICAL TERM
	PREFIX-	-ROOT-	-SUFFIX	
Hypothermia				
Oncolysis				
Primigravida				
Megabladder				

Notes on the slides

- L2 p20–23: slide writes “Pleural (table header)” → **Plural**
- L3 p29: slide writes “Hemi thorax” → **Hemithorax**
- L3 p37: slide writes “Polioencephalities” → **Polioencephalitis**
- L3 p15: slide writes “to loss water” → **to lose water**
- L3 p40: slide writes “Sinistro/o” → **sinistr/o**
- L4 p6: slide writes “Oligouria” → **Oliguria (both accepted)**
- L4 p14: slide writes “Reputure / Rapture” → **Rupture (-rrhexis)**
- L4 p14: slide writes “-Rhage / -Rhea / -Rhexis (headers)” → **-rrhage / -rrhea / -rrhexis**
- L4 p14: slide writes “Diarrhea split “diarr = through, hea = flow”” → **dia = through, -rrhea = flow**
- L5 p8: slide writes “having and intended effect” → **having an intended effect**
- L2 p16: slide writes “Rule numbered 4 twice” → **listed as rule 4b**
- Medical idioms is listed in the course plan for L4, but no slide covers it, so it is not in this pack.

Prepared by Mohammed Aymen · PharmD Clinical 3 · Fall 2026 · Unofficial student summary — the doctors' slides are the reference.